# PROJECT REPORT

# ON

# FAKE NEWS DETECTION MODEL

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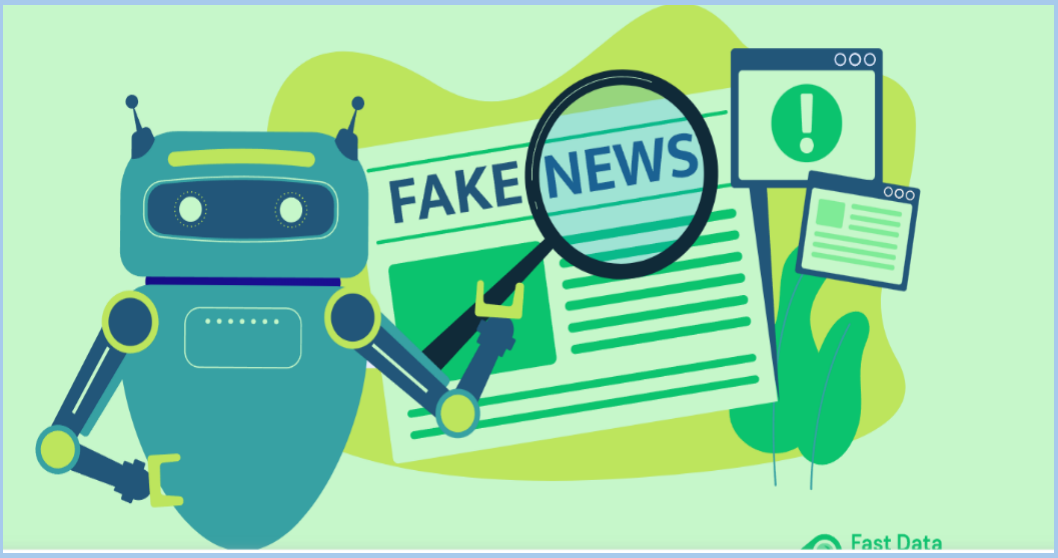
# FAKE NEWS DETECTION MODEL



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# INTRODUCTION

There was a time if anyone required any news ,he or she would sit up for the next day newspaper. With the expansion of online news, today social networking news, online news portal and alternative media became the most sources of report through that fascinating and breaking news are shared at fast pace. Several news portal serve interest by feeding distorted , part correct news to draw in the eye of cluster of individuals. Fake news is information that is not true but it is shared as if it were real. Fake news can be about famous people, politics or even health. The term fake news has become a buzz word lately. Fake news a become a significant concern for being harmful typically spreading confusion and deliberate misinformation among the individuals. Fake news can cause problems because people believe in wrong information. The need for detecting fake news has never been more crucial. Machine learning techniques can help us detect fake news efficiently and accurately. Using natural language processing, machine learning algorithms can accurately detect and categorize true and false news. Detecting fake news using machine learning techniques would require an automatic detection system and it would be trained on dataset containing examples of fake and true news from various sources. This project develops AIML based learning model to automatically classify news as fake or real.

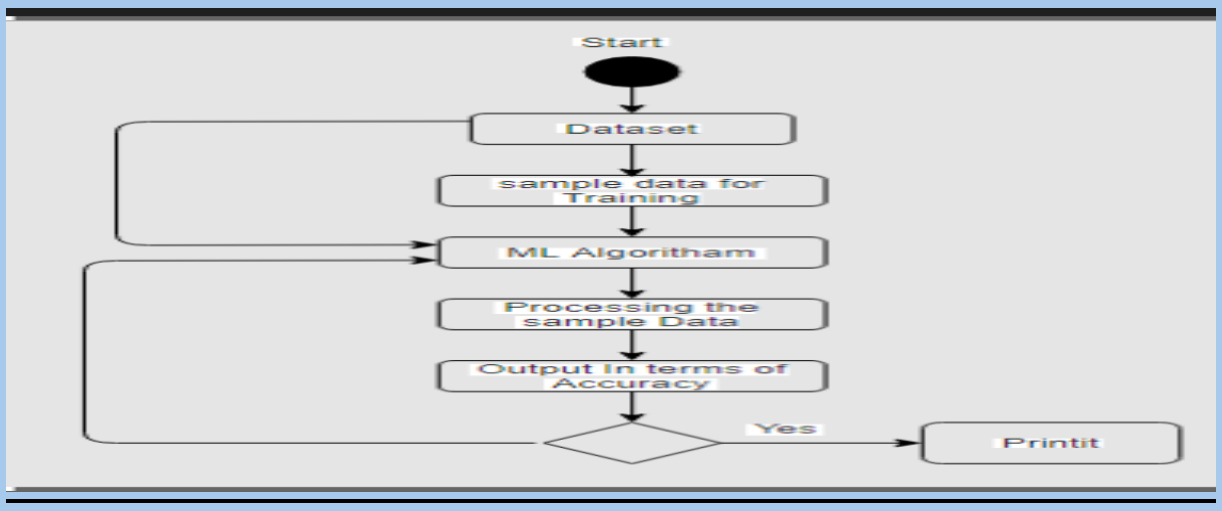
# MOTIVATION OF THE PROJECT

The motivation behind this project stems from the viral spread of misinformation that misleads the public and destabilizes social discourse. Traditional manual verification is too slow to keep pace with the millions of posts generated online creating an urgent need for an automated real time solution. technologically , the motivation lies in exploring the power of online learning algorithms. By developing a system that can categorize a headline as real or fake ,the project aims to empower users with credibility shield,reducing the impact of clickbait and restoring trust in digital information ecosystems.

# PROBLEM DEFINATION

The exponential increase of misleading, false information created with the intention to deceive, causing significant harm to social, economic, and political stability. The project uses logistic regression or passive -aggressive classifier for predicting the news is real or fake, addressing the rapid spread of misinformation on social media. It seeks to solve the challenge of accurately distinguishing deceptive content from legitimate news to restore media credibility and bolster information integrity.

Manual fact-checking is slow, expensive, and ineffective against the rapid, large-scale proliferation of digital misinformation



# OBJECTIVE OF THE PROJECT

The objective of the project is to examine the possible significances related with the spread of fake news. The goal is to develop an automated system utilizing machine learning and natural language processing capable of distinguishing between authentic news and fake news. The project aims to mitigate the rapid spread of misinformation that threats public discourse, social stability and media intergrity. It will implement NLP techniques and would a solution to reduce public confusion and restore trust in digital platforms. As the fake news is a problem that is heavily affecting society and our perception of not only the media but also facts and opinions themselves.AI and ML techniques can be used to solve this problem.

# METHOLOGY AND EXISTING METHODS

The methology for detecting fake news follows structured natural language processing that begins with collecting datasets. Raw text undergoes removing punctuation , converting to lowercase and eliminating common keywords. This text is then converted into numerical vectors and finally the data is split into testing set to build a working model with its performance measured by precision and accuracy.

The existing method and algorithms include logistic regression which is used on classification based on word counts and structure .convolutional neural networks and long short term memory networks are also used to detect complex patterns in text.

# PROS AND CONS OF THE EXISITING METHOD

## Pros of Existing Methods

- **Speed and Scalability:** Automated algorithms can process vast amounts of social media content faster than humans.
- **High Accuracy:** Specific methods like Logistic Regression with feature hashing have shown exceptionally high accuracy
- **Feature Extraction:** Deep Learning models can automatically extract high-dimensional, complex patterns from text, images, and user behaviors without manual feature engineering.
- **Knowledge-Based Verification:** Utilizing external knowledge graphs allows for checking the factual accuracy of claims against trusted sources.

## Cons of Existing Methods

**Lack of Contextual Understanding:** Algorithms often struggle with sarcasm, irony, nuanced half-truths, and cultural context compared to humans.

- Supervised learning methods require enormous amounts of labelled training data, which is often difficult to obtain and may become outdated.
- **Privacy and Transparency Issues:** Many systems lack transparency, making it unclear how they classify content, and they may rely on private user data.

# SCOPE OF THE PROJECT

The scope of fake news detection project defines the technical implementation ,the types of data analyzed and the functional and educational impact. The primary focus is on analyzing the patterns of headlines and to classify them as real or fake. The project focuses on real time classification of news headlines using the passive- aggressive classifier. Unlike the traditional methods ,this method is potential for updates meaning the model can learn from the fake news without needing the entire dataset.

It will help to reduce the viral spread of misinformation by providing a tool for quick verification .

# HARDWARE AND SOFTWARE REQUIREMENTS

1. Hardware requirements

- Processor – intel core i3 and ryzen 3 or equivalent
- Input devices- standard keyboard and mouse
- Display- 1024 and 768 resolution or higher

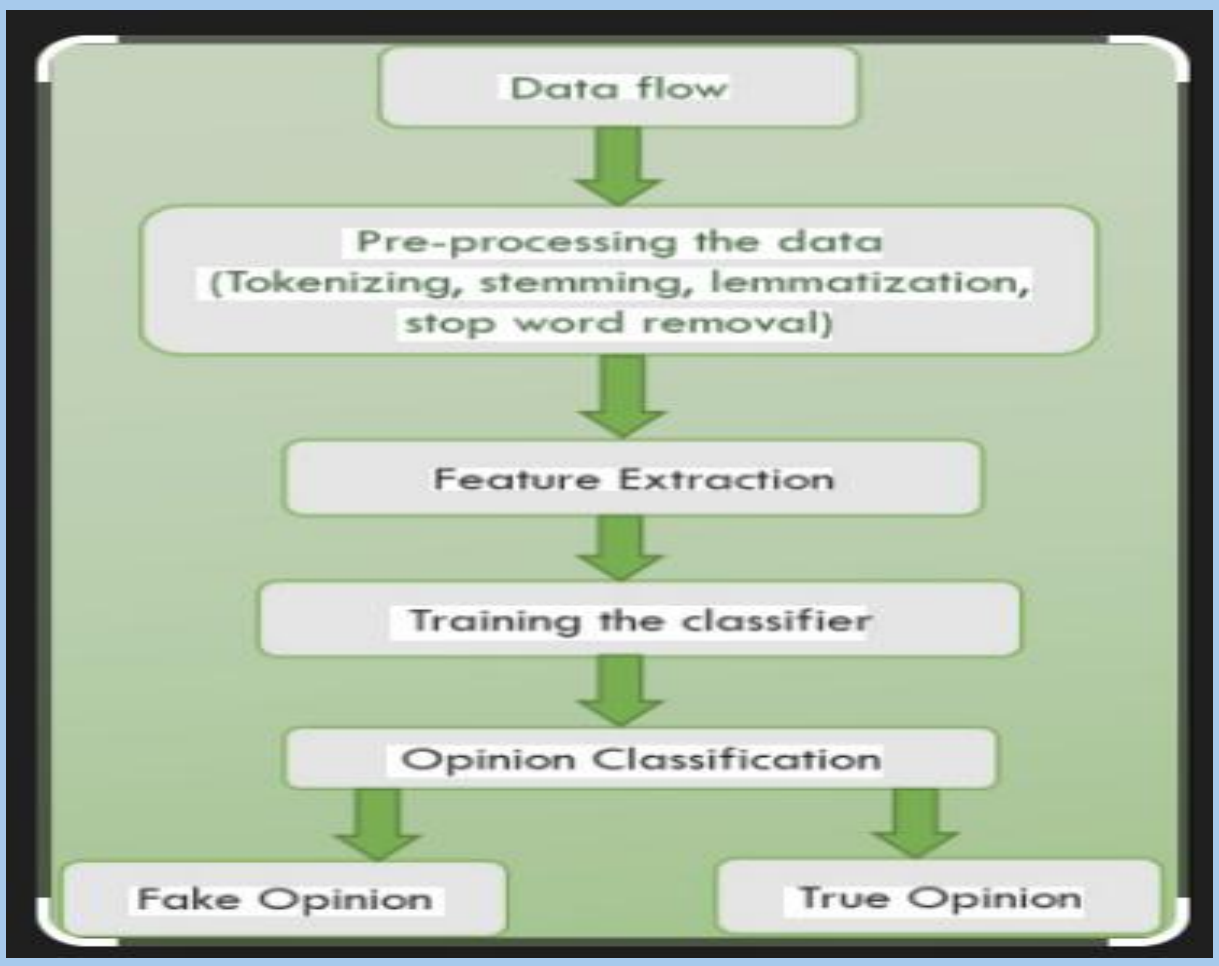
2. Software requirements

- Operating system- windows11
- Programming language- python
- Integrated development environment-vs code or Jupyter notebook.
- Core libraries;-

  math

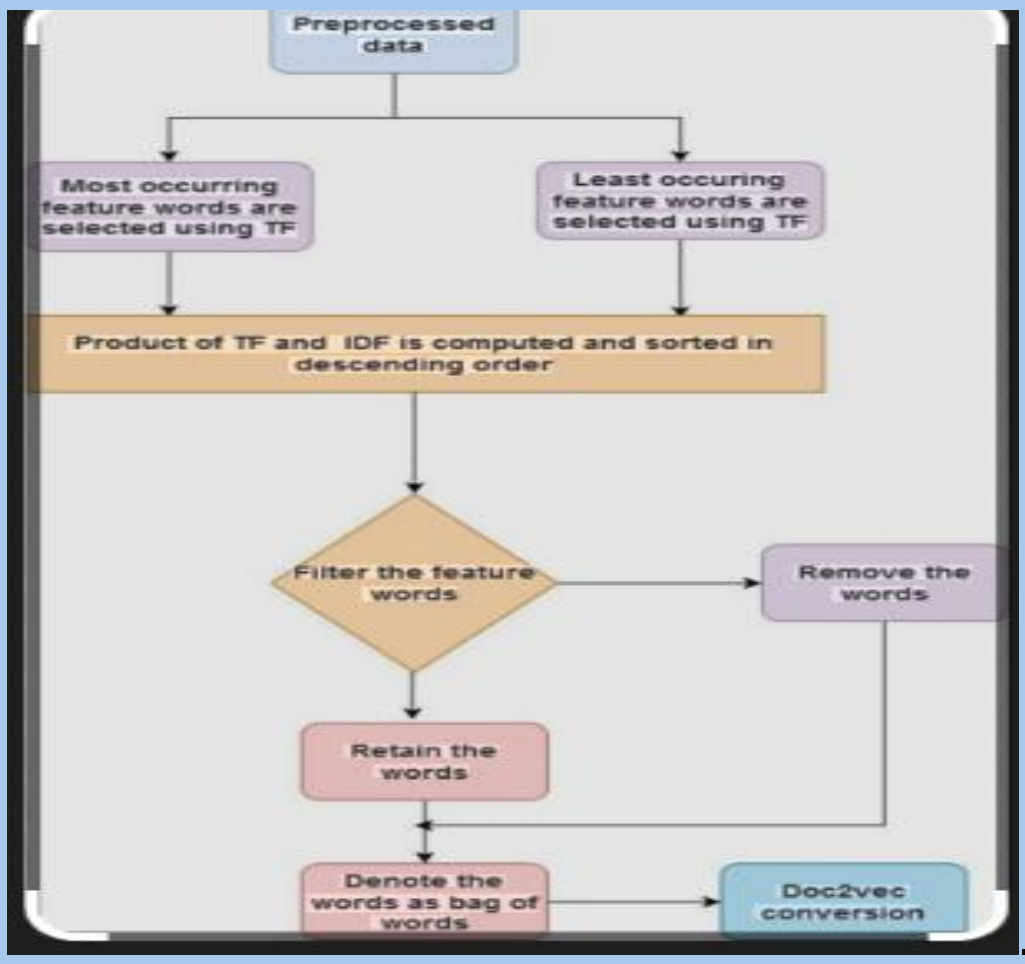
# PROJECT DESIGN AND ARCHETECTURE

This project follows the detection of fake news headlines using the passive-aggressive classifier and follows a modular structure.




- The system follows a layered architecture to ensure clear separation between fake and real news.

- It accepts raw text from user and converts to lowercase to focus on keywords.
- Transforms the data into numerical format using the tf-idf vectorization.
- The core engine PAC analyzes and predict whether the news is real or fake.
- It displays the final prediction.



# ALGORITHM DEVELOPMENT

The core algorithm follows a hinge loss optimization strategy means it process one data point at a time and has two rules-

1. Passive -: If the model correctly predicts the label (Real/Fake) with sufficient confidence, the weights
2. Aggressive- If the prediction is incorrect or lacks a sufficient margin, the model "aggressively" updates the weights to correct the error while staying as close as possible to the previous weight vector.

Since the PAC cannot read the text directly, I use TF-IDF(term frequency – inverse document frequency)

The algorithm uses the hinge loss function to calculate error.

$L=\max(0,1-y.(w.x)$

If L=0,the model is passive

If L>0 the model is aggressive

# CODING AND OUTPUT

The coding focuses on converting text into high dimensional vectors for PAC model.

fake news.py

C: > Users > Sarisha > OneDrive > Documents > Desktop > fake news.py > transform_to_tfidf

```python
import math
import re
# 1 = Real News, 0 = Fake News
data = [
    {"headline": "Scientists discover new planet in habitable zone", "label": 1},
    {"headline": "Government to ban all internet use by 2026", "label": 0},
    {"headline": "Local hero saves cat from burning building", "label": 1},
    {"headline": "Alien invasion confirmed by secret NASA documents", "label": 0},
    {"headline": "Stock market hits record high amid economic growth", "label": 1},
    {"headline": "Drinking bleach cures all known viral diseases", "label": 0},
    {"headline": "New medical breakthrough in cancer research", "label": 1},
    {"headline": "Celebrity caught in massive offshore tax scandal", "label": 1},
    {"headline": "Moon is actually made of cheese, says new study", "label": 0},
    {"headline": "Olympic athlete breaks world record in 100m sprint", "label": 1},
    {"headline": "illegal aliens voting in large numbers in election","label":0}
]
def tokenize(text):
    return re.findall(r'\w+', text.lower())

# Build Vocabulary and calculate IDF
all_headlines = [tokenize(item["headline"]) for item in data]
vocab = sorted(list(set(word for headline in all_headlines for word in headline)))
word_to_idx = {word: i for i, word in enumerate(vocab)}

num_docs = len(all_headlines)
idf = {}
for word in vocab:
    doc_count = sum(1 for headline in all_headlines if word in headline)
    idf[word] = math.log(num_docs / (1 + doc_count))

def transform_to_tfidf(headline):
    tokens = tokenize(headline)
    vector = [0.0] * len(vocab)
    for word in tokens:
        if word in word_to_idx:
            tf = tokens.count(word) / len(tokens)
            vector[word_to_idx[word]] = tf * idf[word]
```

Ln 34, Col 24 Spaces: 4 UTF-8 CRLF {} Python 3.13.7 Python

C: > Users > Sarisha > OneDrive > Documents > Desktop > fake news.py > transform_to_tfidf

```python
def transform_to_tfidf(headline):
    for word in tokens:
        if word in word_to_idx:
            tf = tokens.count(word) / len(tokens)
            vector[word_to_idx[word]] = tf * idf[word]
    return vector

# 3. PASSIVE-AGGRESSIVE CLASSIFIER (PAC) IMPLEMENTATION
class ManualPAC:
    def __init__(self, input_dim, C=1.0):
        self.weights = [0.0] * input_dim
        self.C = C  # Aggressiveness parameter

    def predict_raw(self, x):
        return sum(w * xi for w, xi in zip(self.weights, x))

    def train(self, x, y):
        # y must be 1 or -1 for PAC math
        target = 1 if y == 1 else -1
        prediction = self.predict_raw(x)

        # Calculate Hinge Loss: max(0, 1 - y * (w . x))
        loss = max(0, 1 - target * prediction)

        if loss > 0:
            # Update weights: w = w + (loss / ||x||^2) * y * x
            # Restricted by C (Aggressiveness)
            sq_norm_x = sum(xi**2 for xi in x) or 1e-9
            tau = min(self.C, loss / sq_norm_x)

            for i in range(len(self.weights)):
                self.weights[i] += tau * target * x[i]

    def predict(self, headline):
        x = transform_to_tfidf(headline)
        return "REAL" if self.predict_raw(x) >= 0 else "FAKE"
```

fake news.py

C: > Users > Sarisha > OneDrive > Documents > Desktop > fake news.py > transform_to_tfidf

```python
class ManualPAC:
    def train(self, x, y):
        prediction = self.predict_raw(x)

        # Calculate Hinge Loss: max(0, 1 - y * (w . x))
        loss = max(0, 1 - target * prediction)

        if loss > 0:
            # Update weights: w = w + (loss / ||x||^2) * y * x
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            sq_norm_x = sum(xi**2 for xi in x) or 1e-9
            tau = min(self.C, loss / sq_norm_x)

            for i in range(len(self.weights)):
                self.weights[i] += tau * target * x[i]

    def predict(self, headline):
        x = transform_to_tfidf(headline)
        return "REAL" if self.predict_raw(x) >= 0 else "FAKE"

# 4. RUNNING THE MODEL
model = ManualPAC(len(vocab), C=1.0)
for item in data:
    vec = transform_to_tfidf(item["headline"])
    model.train(vec, item["label"])
print("--- Fake News Detector (PAC from Scratch) ---")
test_headline = input("Enter a news headline to check: ")
result = model.predict(test_headline)
print(f"Prediction: {result}")
```
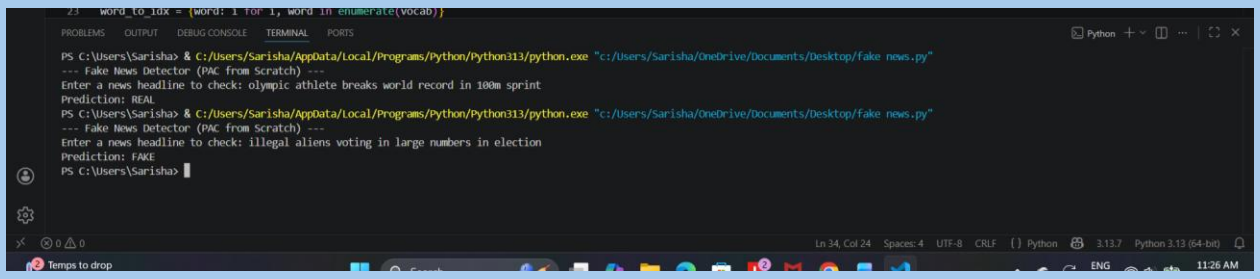

# Output and testing:-




The developed model is highly accurate and efficient .

Significant outcomes of the model:-

The final script is extremely light weight and can run on low power hardware.

## PROJECT APPLICABLILITY ON REAL -WORLD APPLICATIONS

- The real -world applicability of this system lies in its ability to serve as a gatekeeper for digital platforms.it is uniquely suited for the modern news cycle, allowing the system to adapt to new tactics without requiring time -consuming process.
- Platforms like twitter and facebook can use PAC models to scan headlines instantly as they are posted.
- Large news agencies can use the system to filter out the fabricated contents.
- The model can serve an early detection tool, alerting newsroom to a surge in suspicious headlines before they gain public traction.
- It acts as a specialized spam filter for information removing noise and intentionally misleading headlines.
- During crises like a pandemic ,the model can be deployed to suppress false health advice or rumors that could lead to physical harm.

# .LIMITATIONS OF THE SYSTEM AND THE CHALLENGES FACED IN THE MAKING OF THE PROJECT

- Lack of visual analysis- the system is text -blind to images.it cannot detect if a real headline is paired with a manipulated image.
- The system lacks full context found in complete articles.
- The TF-IDF vectorizer can become outdated if the vocabulary isn't periodically refreshed.

Challenges faced-

- Setting up the python environment for libraries was a time consuming task.
- It was difficult to find the balanced datasheet specifically for headlines.

# CONCLUSIONS AND FUTURE ENHANCEMENTS

- The project successfully demonstrates that a passive - aggressive classifier is a highly efficient solution for detecting fake news. By achieving high accuracy rates, the model proves that even short form text contains markers to distinguish misinformation. By accurately classifying misinformation with minimal latency,this system offers a practical tool for curbing the viral spread of digital wildfires and promoting a more responsible information ecosystem.

For future enhancements , I would like to Move beyond TF-IDF to contextual word embeddings and deep learning models .I would like to Integrate computer vision (CNNs) to analyze images, videos, and memes attached to news stories, as fake news often uses manipulated visuals, moving beyond text-only analysis .also I would Expand the model to detect fake news in languages other than English and ensure the model generalizes across different domains (e.g., politics, health, technology) rather than overfitting to one dataset.

# REFERENCES


- Datasets –
  Kaggle fake news dataset-www.kaggle.com
- H. Ahmed, I. Traore, and S. Saad, "Detecting opinion spams and fake news using text classification," *Journal of Security and Privacy*, 2018.
- RK Kaliyar, A Goswami, P Narang, "Fake BERT: Fake news detection in social media with a BERT-based deep learning approach,".
- "Multiple features based approach for automatic fake news detection on social networks using deep learning," SR Sahoo, BB Gupta.
- **TF-IDF Vectorization** (Term Frequency-Inverse Document Frequency) for text conversion.
- **Logistic Regression / Passive-Aggressive Classifier** for classification.
- **NLP Techniques** (Tokenization, Punctuation Removal)

# THANK YOU